Yel Scrypt Fire Studio

Low Poly Road Vehicles Free Package

Assets Documentation

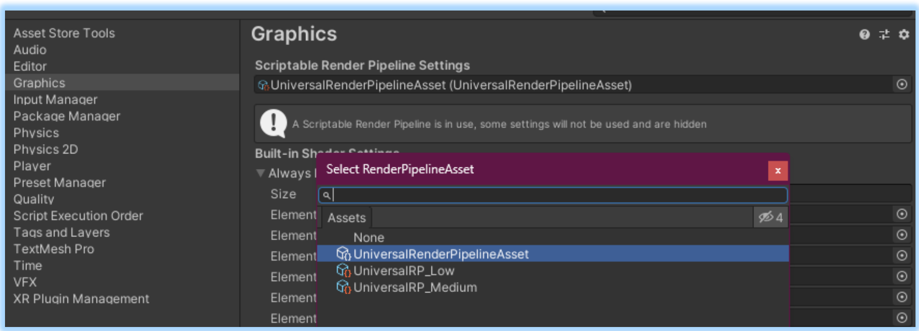
Version 1.1.0

CONTENIDO

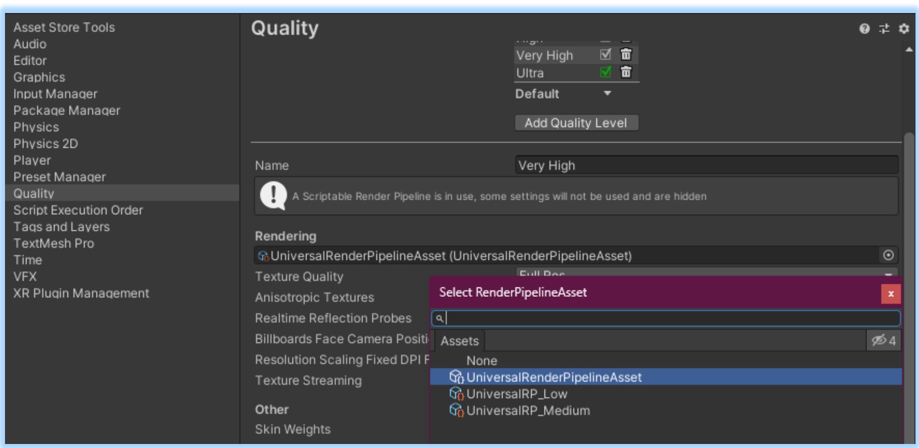
HOW TO ACTIVATE UNIVERSALRP

*This step is not needed if your project is configured in URP by default. *

- (Optional – To create the URP assets) In the **Project** window, **right click > Create > Rendering > URP Asset (with Universal Renderer)**, and move it in a convenient folder.
- Go to **Edit > Project Settings**
- In the configuration windows, go to **Graphics** and set the URP asset, clicking on the circle and selecting the asset.



- Go to **Quality** and set the same URP asset.

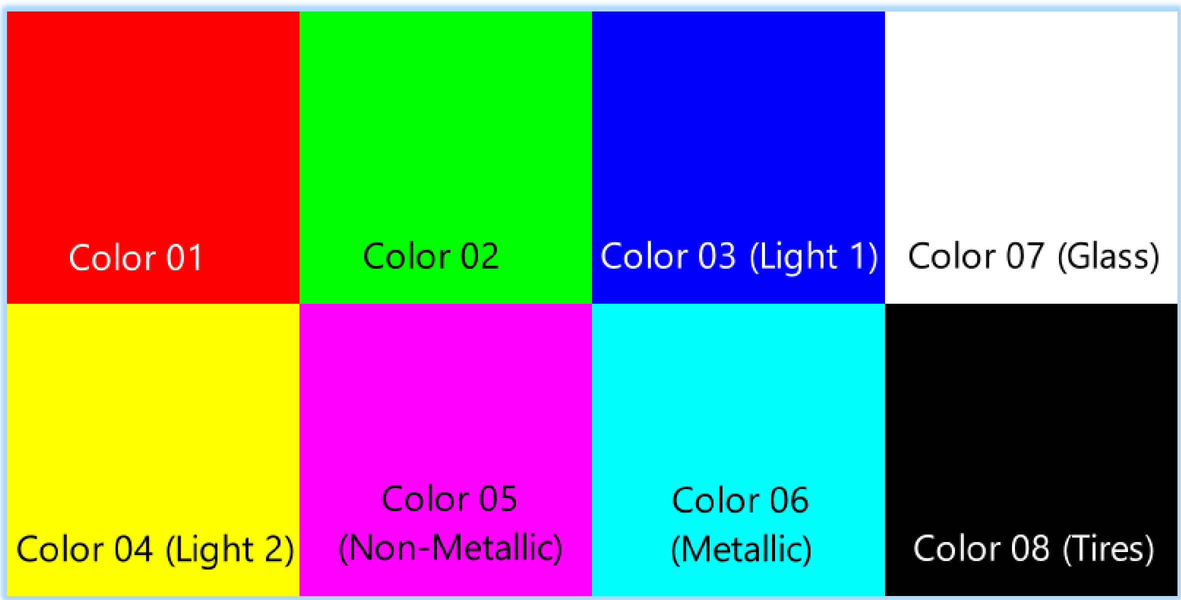


- **If files are not recognized, please reimport the URP assets.**
- To change from URP to Bult-in, just select **None** instead of the URP asset.

MESHES AND UV MAPPING

- 10 3D models (FBX) and 10 prefabs.
- Each prefab has an extra object with the colliders.
- One decal texture (1024x1024px), and a [main texture for color (8x4px).]
- 2 UV maps:
  - UV0: Low poly 8 islands.
  - UV1: unwrapped for decal textures.
- Not rigged.
- You can find the UV layouts of UV1 in folder: Assets>YelScryptFireStudio>LowPolyRoadVehiclesPart1>Doc>UV_Layouts
- Lit materials with Metallic Workflow (No PBR treatment)

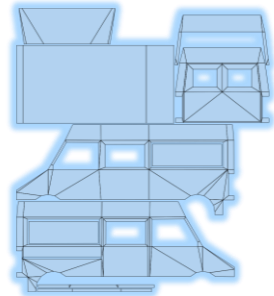
MAIN TEXTURE FOR COLOR MASK



AMBULANCE_01

Poly count: 4152 tris
Meshes (inside FBX): 9
Measures:

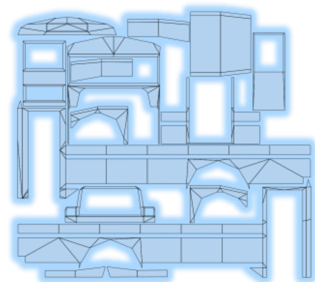
- X: 2.13 m
- Y: 2 m
- Z. 4.23 m



BUS_01

Poly count: 7068 tris
Meshes (inside FBX): 8
Measures:

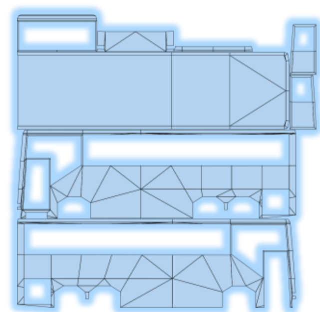
- X: 1.85 m
- Y: 2.09 m
- Z: 6.34 m



BUS_02

Poly count: 8224 tris
Meshes (inside FBX): 10
Measures:

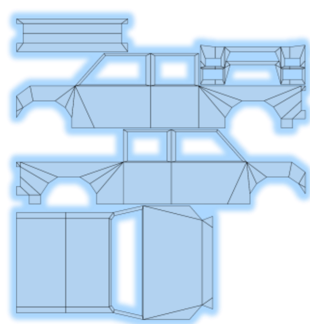
- X: 1.84 m
- Y: 2.21 m
- Z: 6.33 m



PICKUP_01

Poly count: 5986 tris
Meshes (inside FBX): 10
Measures:

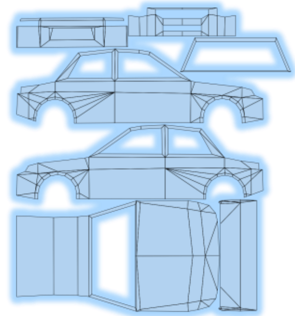
- X: 2.14 m
- Y: 1.75 m
- Z: 4.95 m



SEDAN_01

Poly count: 3136 tris
Meshes (inside FBX): 8
Measure:

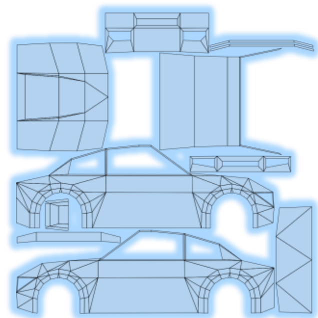
- X: 1.95 m
- Y: 1.25 m
- Z: 3.88 m



SPORTCAR_01

Poly count: 3306 tris
Meshes (inside FBX): 8
Measures:

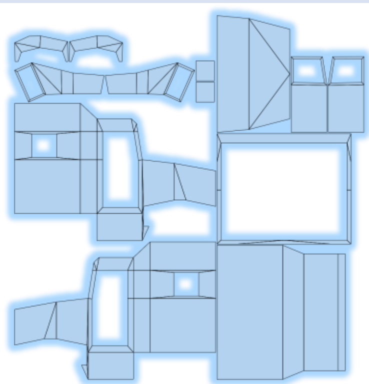
- X: 2.04 m
- Y: 1.31 m
- Z: 3.98 m



TRUCK_01

Poly count: 10086 tris
Meshes (inside FBX): 10
Measures:

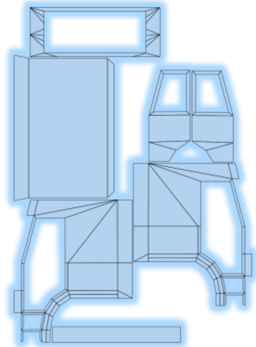
- X: 2.25 m
- Y: 2.71 m
- Z: 4.54 m



TRUCK_02

Poly count: 6496 tris
Meshes (inside FBX): 8
Measures:

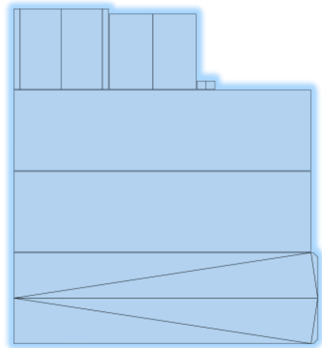
- X: 2.22 m
- Y: 2.25 m
- Z: 3.37 m



TRUCKCONTAINER_01

Poly count: 5360 tris
Meshes (inside FBX): 6
Measures:

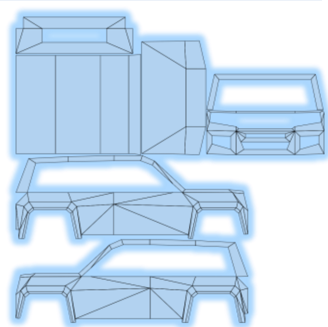
- X: 1.99 m
- Y: 2.47 m
- Z: 6.37 m



VAN_01

Poly count: 3776 tris
Meshes (inside FBX): 8
Measures:

- X: 2.25 m
- Y: 1.52 m
- Z: 3.68 m

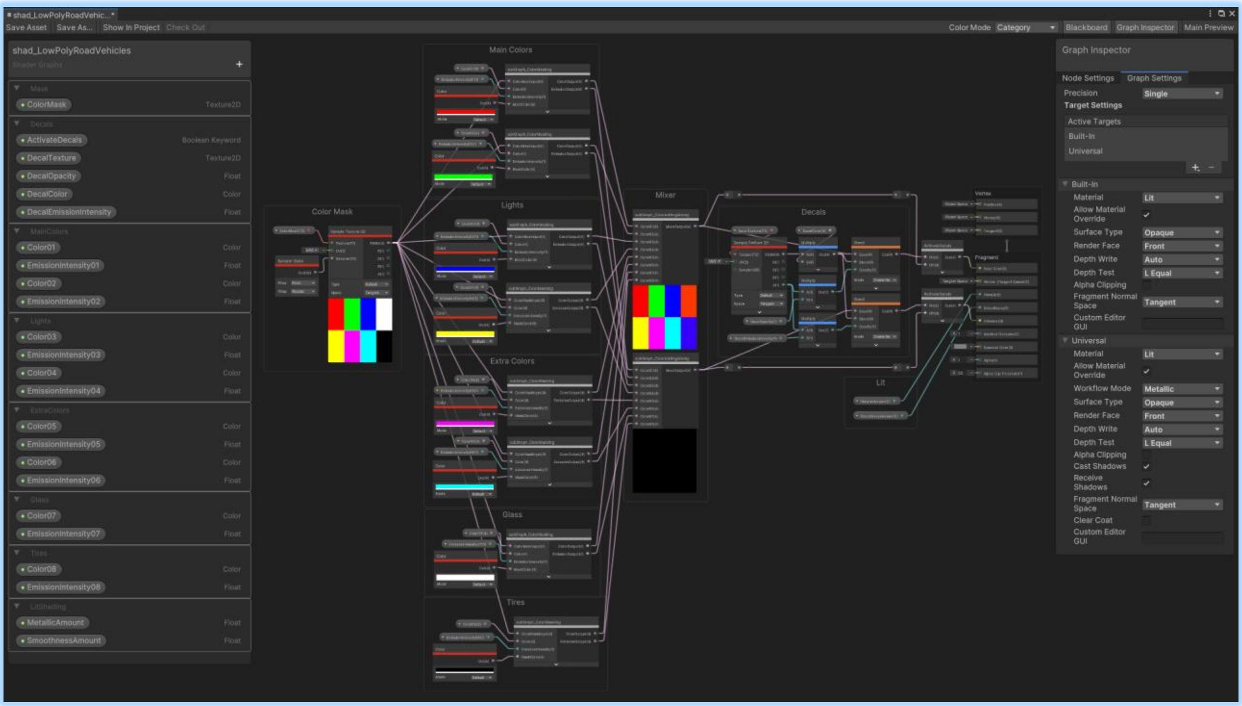


LOW POLY ROAD VEHICLES SHADER GRAPH

Shader for low poly meshes, using color masking.

Takes a texture divided by specific colors. Using several color masks, this texture can be used for generate up to 8 customizable colors, categorized by sections:

Sections: **Color Mask, Color Sections (Main Colors, Lights, Extra Colors, Glass, Tires), Mixer, Decals, Lit and ActivateDecals Boolean.**



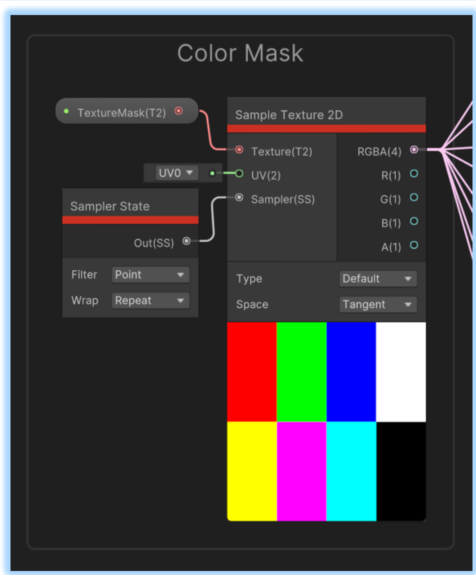
COLOR MASK

The input of [a special texture of 8 discrete colors](). Each color acts as a mask guide to isolate each section with its UV island and allow changing color individually.

UV map 0 from the meshes is used.

Sampler State is included to ensure that interpolation is not used in this mask.

Output: Color Masking Sections



COLOR SECTIONS

Group of sections that accomplish the color isolation and color changing. Also, they include emission conversion.

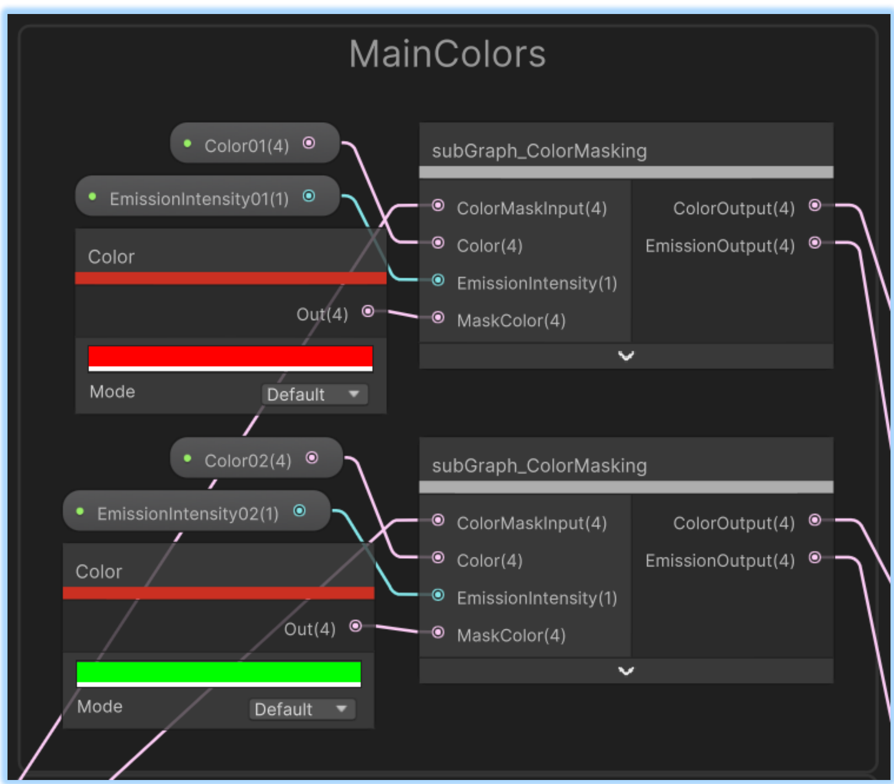
The **Color** variable is for select the output color of its corresponding section of the color mask. **Color03** and **Color04** are the only ones that are in **HDR mode**.

The **EmissionIntensity** variable indicates the amount of output color is passed to the **Emission** of the material.

The **MaskColor** variable indicates which section of the **ColorMask** is used for coloring, by matching the corresponding color.

Input: Texture RGBA → ColorMaskInput

Output: ColorOutput → ColorAddingMixing SG (1)
EmissionOutput → ColorAddingMixing SG (2)



MIXER

Take the outputs from all **Color Sections Sub-graphs** to mixing back into a single color-matrix.

The **ColorAddingMixing SG** is used 2 times, one for color and the other for emission.

Input: ColorOutput → ColorAddingMixing SG (1)
EmissionOutput → ColorAddingMixing SG (2)

Output: ColorAddingMixing SG (1) → Decals
→ActivateDecals (1)

ColorAddingMixing SG (2) → Decals
→ActivateDecals (2)

DECALS

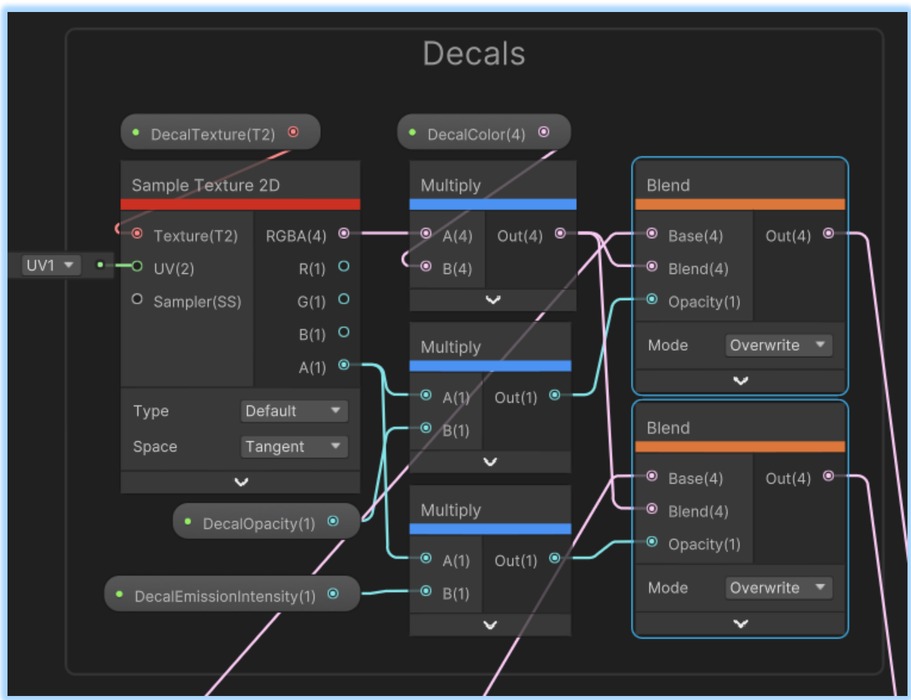
It takes and combine a decal texture over the colors. It uses the UV map 1 from the meshes. Greyscale and transparent textures are recommended.

First, set its color by Multiply, then add the decal over the colors by **Overwrite blending**, using the **Alpha channel** for the **Opacity**. In both cases, the **Alpha channel** is regulated by **DecalOpacity** and **DecalEmissionIntensity** respectively.

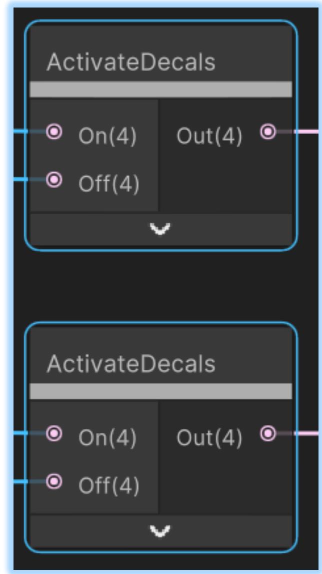
*Remember set **DecalColor** as white if your decal texture is colored, otherwise the colors of your decal texture will be changed.

Input: ColorAddingMixing SG (1) → DecalOpacity's Base Blend
ColorAddingMixing SG (2) → DecalEmissionIntensity's Base Blend

Output: DecalOpacity's Blend → ActivateDecals (1)
DecalEmissionIntensity's Blend → ActivateDecals (2)



BOOLEANS



ActivateDecals Boolean activates the use of the decal texture. The switching is at the last step of the shader graph.

On > with decals.

Off > Without decals.

Input: ColorAddingMixing SG (1) → Off (1)
Decal's Blend (1) → On (1)
ColorAddingMixing SG (2) → Off (2)
Decal's Blend (2) → On (2)

Output: ActivateDecals (1) → Material Base Color
ActivateDecals (2) → Material Emission

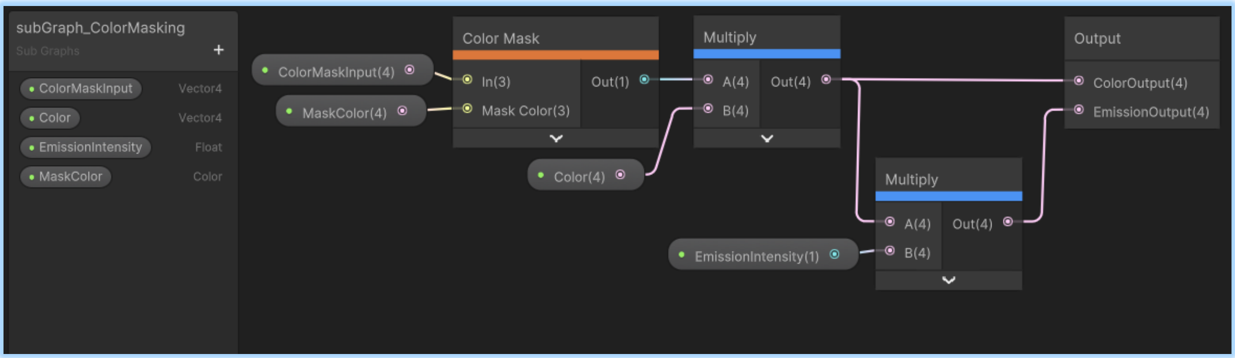
ON INSPECTOR

TextureMask: texture that works as base for color masking.
ActivateDecals: Boolean for activate the decal texture.
DecalTexture: texture for the decals. It includes its Tiling and Offset Input.
DecalOpacity: the opacity of the entire shader. ***(0 <-> 1)***
DecalColor: color for the decal textures. Greyscale textures are recommended to work properly. **HDR** enabled.
DecalEmissionIntensity: the intensity of the emission from the decals. ***(0 <-> 2)***
Color01-08: each of the output colors. **Color03** and **Color04** have **HDR** enabled.
EmissionIntensity01-08: the intensity of the emission from the decals. ***(0 <-> 2)***
MetallicAmount: the amount of metallic shading. ***(0 <-> 1)***
SmoothnessAmount: the amount of smoothness shading. ***(0 <-> 1)***

SUB GRAPHS

COLORMASKING SUB GRAPH

Takes the ColorMaskInput to isolate one of the colors with the MaskColor. Then, the output (as a Pure-White color) is tinted with Multiply by Color. Another Multiply is applied for EmissionIntensity to send it to Emission.



COLORADDINGMIXIG SUB GRAPH

Consist in a group of nested Add nodes in a branching style. Mixing as 8 > 4 > 2 > 1 arrangement.